

RAUNAQ MAKKAR

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Education

Manipal University Jaipur

2022-2026

B.Tech. Computer and Communication Engineering

CGPA: 8.76

Delhi Public School Ghaziabad International

2019-2022

10th Boards: 95%, 12th Boards: 86.8%

Projects

AI-Driven Career Guidance Platform (CareerEdge) | React.js, MongoDB, JWT, Postman | [Link](#)

2025

- Built **CareerEdge**, an AI-powered career guidance platform with secure **JWT-based** role authentication.
- Developed an AI-driven mentor-mentee matchmaking engine using **Jaccard** and **TF-IDF** similarity.
- Integrated a Gemini/GPT API chatbot for real-time, intelligent career guidance.
- Live Demo (Deployed on Vercel): career-edge-frontend.vercel.app

AI-Powered Road Maintenance System (RoadNetAI) | YOLOv8, Python, MongoDB, Cloudinary | [Link](#)

2026

- Developed RoadNet.AI, an AI-powered road infrastructure monitoring system using YOLOv8m-seg trained on 7000+ annotated images for detecting potholes, road cracks, manholes, and open manholes, achieving $mAP@50 = 0.84$, $mAP@50-95 = 0.71$, and $F1-score = 0.82$.
- Built an end-to-end real-time video processing pipeline using OpenCV and FastAPI for frame extraction, segmentation inference, confidence filtering, and annotated defect visualization from vehicle-mounted road inspection videos.
- Implemented a custom Repair Priority Score (RPS) mechanism using weighted severity, confidence, frequency, and duration analysis to intelligently prioritize road maintenance operations and automate ticket generation.
- Designed a scalable backend architecture integrating MongoDB, REST APIs, analytics storage, and dashboard-ready outputs for defect tracking, maintenance monitoring, and smart-city infrastructure analytics.

Experience

Deloitte USI Capstone Program (Winner)

2025

AI-Powered Video Surveillance System

- Built an AI-powered video surveillance system combining **YOLO** weapon detection and **ConvLSTM** violence recognition for real-time alerts and visual threat evidence.
- Developed a YOLOv11 model for weapon detection with 92% precision, real-time inference, and annotated frames for supervisor review.
- Implemented a ConvLSTM violence detector with MobileNet/EfficientNet features, achieving 88% recall and visualizing violent actions.
- Collaborated with Deloitte mentors to refine KPIs (accuracy, response time, false positives) through expert feedback and iterative validation for public safety deployment.

Technical Skills

Programming Languages: C, C++, HTML, CSS, JavaScript, Python

Frameworks & Tools: React.js, Git, GitHub, Postman

Databases: MySQL, MongoDB

Deployment & Authentication: Vercel, Render & JWT

Achievements

Dean's List Award – Manipal University Jaipur: Recognized for consistent academic excellence and high academic standing during the semesters.

Event and Management Team – ACM,MUJ: Led the planning and execution of multiple technical and cultural events, enhancing member engagement and organizational visibility.